

HEALTHCONNECT EXPERIENCE LAB

Final Analytics & Decision Support Report

Track: Data Analytics

Week 8

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Executive Summary

The Week 8 Data Analytics contribution consolidates the HealthConnect appointment analysis, final KPIs, validated findings, business insights, recommendations, testing evidence and cross-track contribution. It builds on the Week 5 baseline, Week 6 advanced analysis and Week 7 testing/refinement. The central validated finding is that previous no-show history is associated with higher observed no-show rates. The analytics work provides an evidence base for operational decision-making and potential predictive modelling.

Project Context and Objective

HealthConnect Clinic is an appointment-based healthcare project focused on reducing missed appointments and improving patient support. The central project question is:

How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

The Data Analytics track addressed this through data preparation, KPI development, Power BI dashboarding, segment-level analysis, validation and decision-support recommendations.

Final Analytics Contribution

- Prepared and analysed the HealthConnect appointment and patient dataset.
- Developed and reviewed key performance indicators for appointment outcomes and operations.
- Built and refined Power BI dashboards using Power Query and DAX.
- Investigated no-show patterns across previous no-show history, distance, reminder status, appointment type and other relevant segments.
- Validated important analytical outputs during Week 7 testing.
- Translated observed patterns into business insights and evidence-informed recommendations.
- Provided analytical evidence to Project Management and relevant analytical input to Data Science.

Final KPIs

KPI	Final Value	Meaning
No-Show Rate	48.5%	Share of appointments recorded as no-shows.
Cancellation Rate	5.3%	Share of appointments recorded as cancelled.
Reminder Coverage	72.7%	Share of appointments with a recorded reminder channel.
Average Waiting Time	24.19 minutes	Average recorded waiting time for appointments with available data.

Key Analytical Findings

The strongest validated finding concerned previous no-show history. The observed no-show rate increased across the practical history groups:

Previous No-Show Group	Appointments	No-Shows	Observed No-Show Rate
0	2,921	1,271	43.5%
1–2	1,986	1,088	54.8%
3+	93	64	68.8%
Total	5,000	2,423	48.5%

Other observed patterns included a higher no-show rate among appointments in the 21+ km distance group and among appointments without a recorded reminder. Follow-up appointments also recorded the highest appointment-type no-show rate in the Week 6 analysis. These are associations in the dataset, not proof of causation.

Testing and Validation

Week 7 focused on testing and refinement rather than creating a new dashboard. The main cross-track test checked the No-Show Rate by Previous No-Show Group against underlying appointment and no-show counts.

Tested Component	Reference Result	Actual Result	Outcome
Previous No-Show Group rates	43.5%, 54.8%, 68.8%	43.5%, 54.8%, 68.8%	PASS
Overall totals	5,000 appointments; 2,423 no-shows	5,000 appointments; 2,423 no-shows	Consistent
Distance/reminder checks	Consistent with Week 6 outputs	No conflicting result identified	Supported

No numerical correction was required. The main refinement was strengthening validation evidence and documentation so supported findings could be carried forward without overstating the scope of testing.

Business Insights and Recommendations

Observed Insight	Decision-Support Implication	Suggested Action
Previous no-show history was associated with higher observed no-show rates.	Previous attendance behaviour may identify groups needing additional support.	Consider targeted confirmation or follow-up workflows for patients with previous no-shows.
The 21+ km group had an elevated observed no-show rate.	Distance may indicate an accessibility or travel-related issue worth investigating.	Explore targeted scheduling, confirmation or support options.
Appointments without a recorded reminder had a higher observed no-show rate than appointments with a reminder.	Reminder coverage is a practical intervention area.	Test broader reminder coverage and segment-specific escalation.

Follow-up appointments had the highest appointment-type no-show rate.	Follow-up appointments may require additional engagement.	Test targeted confirmation and follow-up communication.
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These recommendations are evidence-informed and should be tested operationally before being treated as causal or universally effective.

Data Science Integration

Analytical patterns were shared with the Data Science track around booking lead time, previous no-show behaviour and appointment type $\times$ reminder channel. Data Science proposed potential model features including:

- high_lead_time_flag for booking lead days greater than 30.
- chronic_no_show_flag for patients with two or more previous no-shows.
- An interaction term between appointment type and reminder channel.

The Data Science figures were kept separate from the Week 7 validated figures and were not silently merged. This is important because the analyses used different groupings or processing assumptions. Booking lead-time data had also been identified as having quality concerns, so it was not treated as a fully validated predictive feature without further reconciliation.

Project Management Integration

Data Analytics provided validated no-show analysis and testing evidence to Project Management. This supported testing documentation, evidence-based readiness tracking and project coordination. The verified cross-track activity was the validation of the No-Show Rate by Previous No-Show Group; expected and actual rates matched.

Final Outcome

- Final high-level KPIs were consolidated into the analytics decision-support package.
- The key previous-no-show finding was validated against underlying counts and calculations.
- Business insights and evidence-informed recommendations were documented.
- Cross-track analytics evidence was provided to Project Management and relevant analytical input was shared with Data Science.
- The analytics contribution was prepared for inclusion in the wider HealthConnect solution and final presentation.

Analytical Limitations

- The analysis identifies observed associations and differences; it does not establish causation.
- The 3+ previous no-show group contains relatively few appointments and should be interpreted cautiously.
- Some dataset variables contain missing values, including waiting time and distance.
- Booking lead-time data requires quality review before being treated as a validated predictive feature.
- The analytics evidence is component-level; complete project-level end-to-end validation depends on the wider HealthConnect workstreams.
- Recommendations should be tested in practice before effectiveness is assumed.

Final Integration Contribution and Presentation

The Data Analytics contribution acts as an evidence layer for the HealthConnect solution. It provides measured appointment outcomes, validated patterns, business insights and recommendations that other workstreams can consider when developing predictive and operational components.

Effective analytics is not only about building a dashboard. It involves preparing the data, measuring performance, identifying meaningful patterns, testing the results, understanding limitations and communicating findings in a way that supports responsible decisions.

Conclusion

The HealthConnect Data Analytics track created a tested analytical contribution consisting of validated KPIs, documented findings, business insights, recommendations and cross-track evidence. The next step is to combine this evidence with the wider predictive, ML, GenAI and project-coordination work so the complete HealthConnect solution can be demonstrated coherently.